
TIME SERIES

DEFINITION

The arrangement of data in accordance with their time of occurrence is a time series. It is the chronological arrangement of data. Here, time is just a way in which one can relate the entire phenomenon to suitable reference points. Time can be hours, days, months or years.

A time series depicts the relationship between two variables. Time is one of those variables and the second is any quantitative variable. It is not necessary that the relationship always shows increment in the change of the variable with reference to time. The relation is not always decreasing too.

It may be increasing for some and decreasing for some points in time. Can you think of any such example? The temperature of a particular city in a particular week or a month is one of those examples.

Uses of Time Series:

- The most important use of studying time series is that it helps us to predict the future behaviour of the variable based on past experience
- It is helpful for business planning as it helps in comparing the actual current performance with the expected one
- From time series, we get to study the past behaviour of the phenomenon or the variable under consideration
- We can compare the changes in the values of different variables at different times or places, etc.

COMPONENTS OF TIME SERIES

I. Secular Trend

II. Seasonal Variations

III. Cyclical Variations

IV. Irregular Variations

1. Trend:

The trend shows the general tendency of the data to increase or decrease during a long period of time. A trend is a smooth, general, long-term, average tendency. It is not always necessary that the increase or decrease is in the same direction throughout the given period of time.

It is observable that the tendencies may increase, decrease or are stable in different sections of time. But the overall trend must be upward, downward or stable. The population, agricultural production, items manufactured, number of births and deaths, number of industry or any factory, number of schools or colleges are some of its example showing some kind of tendencies of movement.

Linear and Non-Linear Trend :

If we plot the time series values on a graph in accordance with time t . The pattern of the data clustering shows the type of trend. If the set of data cluster more or less round a straight line, then the trend is linear otherwise it is non-linear (Curvilinear).

2. Seasonal Variations:

These are the rhythmic forces which operate in a regular and periodic manner over a span of less than a year. They have the same or almost the same pattern during a period of 12 months. This variation will be present in a time series if the data are recorded hourly, daily, weekly, quarterly, or monthly.

These variations come into play either because of the natural forces or man-made conventions. The various seasons or climatic conditions play an important role in seasonal variations. Such as production of crops depends on seasons, the sale of umbrella and raincoats in the rainy season, and the sale of electric fans and A.C. shoots up in summer seasons.

The effect of man-made conventions such as some festivals, customs, habits, fashions, and some occasions like marriage is easily noticeable. They recur themselves year after year. An upswing in a season should not be taken as an indicator of better business conditions.

3. Cyclic Variations:

The variations in a time series which operate themselves over a span of more than one year are the cyclic variations. This oscillatory movement has a period of oscillation of more than a year. One complete period is a cycle. This cyclic movement is sometimes called the 'Business Cycle'.

It is a four-phase cycle comprising of the phases of prosperity, recession, depression, and recovery. The cyclic variation may be regular or not periodic. The upswings and the downswings in business depend upon the joint nature of the economic forces and the interaction between them.

Irregular Variations

Random variations are fluctuations which are a result of unforeseen and unpredictable forces. These forces operate in an absolutely random or erratic manner and do not have any definite pattern. Thus, these variations may be due to floods, famines, earthquakes, strikes, etc. It is often caused by random or unforeseeable events that cannot be easily attributed to specific factors or variables.

Example:

When a powerful storm, like a hurricane or a thunderstorm, passes through the city, it can lead to rapid and unpredictable fluctuations in temperature. For example, a sudden drop in temperature during a summer thunderstorm is irregular variation in the time series data.

These irregular variations in temperature data occur due to specific, transient events like weather patterns, storms, or atmospheric conditions that temporarily disrupt the typical seasonal pattern.

MODELS OF TIME SERIES

The following are the two models which we generally use for the decomposition of time series into its four components. The objective is to estimate and separate the four types of variations and to bring out the relative effect of each on the overall behavior of the time series.

1. Additive Model
2. Multiplicative Model

1) Additive Model

In the additive model, we represent a particular observation in a time series as the sum of these four components.

$$Y(t) = T(t) + S(t) + C(t) + I(t)$$

i.e. $Y = T + S + C + I$

Where

$Y(t)$ = Value of the variable at time t

$T(t)$ = Trend Value at time t

$S(t)$ = Seasonal Variation at time t

$C(t)$ = Cyclic Variation at time t

$L(t)$ = Irregular or Random Variation at time t

In another way, we can write i.e. $Y = T + S + C + I$

2) Multiplicative Model

In this model, four components have a multiplicative relationship. So, we represent a particular observation in a time series as the product of these four components:

$$Y(t) = T(t) \times S(t) \times C(t) \times I(t)$$

i.e. $Y = T \times S \times C \times I$

where Y, T, S, C and I represents the terms as in additive model.

In another way, we can write $Y = T \times S \times C \times I$

This model is the most used model in the decomposition of time series. To remove any doubt between the two models, it should be made clear that in Multiplicative model S, C, and I are indices expressed as decimal percentages whereas, in Additive model S, C and I are quantitative deviations about a trend that can be expressed as seasonal, cyclical and irregular in nature.

Examples

If in a multiplicative model.

$T = 500$, $S = 1.4$, $C = 1.20$ and $I = 0.7$

then $O = T \times S \times C \times I$

By substituting the values we get

$$O = 500 \times 1.4 \times 1.20 \times 0.7 = 588$$

If in additive model,

$T = 500$, $S = 100$, $C = 25$, $I = -60$

then $O = 500 + 100 + 25 - 60 = 565$

ESTIMATION OF TREND

1. Freehand or Graphic Method:

It is the simplest and most flexible method for estimating a trend. We will see the working procedure of this method.

Procedure:

- (a) Plot the time series data on a graph.
- (b) Draw a freehand smooth curve joining the plotted points.
- (c) Examine the direction of the trend based on the plotted points.
- (d) Draw a straight line which will pass through the maximum number of plotted points.

2. Method of Semi-Averages

In this method, the semi-averages are calculated to find out the trend values. Now, we will see the working procedure of this method.

Procedure:

- (i) The data is divided into two equal parts. In case of odd number of data, two equal parts can be made simply by omitting the middle year.
- (ii) The average of each part is calculated, thus we get two points.
- (iii) Each point is plotted at the mid-point (year) of each half.
- (iv) Join the two points by a straight line.
- (v) The straight line can be extended on either side.
- (vi) This line is the trend line by the methods of semi-averages.

Example 1: Fit a trend line by the method of semi-averages for the given data.

+	Year	2000	2001	2002	2003	2004	2005	2006
	Sale	105	115	120	100	110	125	135

Solution:

Since the number of years is odd(seven), we will leave the middle year's production value and obtain the averages of first three years and last three years.

+	Year	Sale	Average
	2000	105	$\frac{105 + 115 + 120}{3} = 113.33$
	2001	115	
	2002	120	
	2003	100(left out)	
	2004	110	$\frac{110 + 125 + 135}{3} = 123.33$
	2005	125	
	2006	135	

(i) Fit a trend line to the following data by method of semi-averages.

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007
Production (in thousand units)	12	15	20	18	25	32	30	40	44

(ii) Also, find 3-yearly moving averages and draw these on a graph paper along with the original time series.

Solution :

(i) Since data is available for $n = 9$ (odd) years, the two parts will consist of years 1999 to 2002 and 2004 to 2007, the year 2003 being omitted.

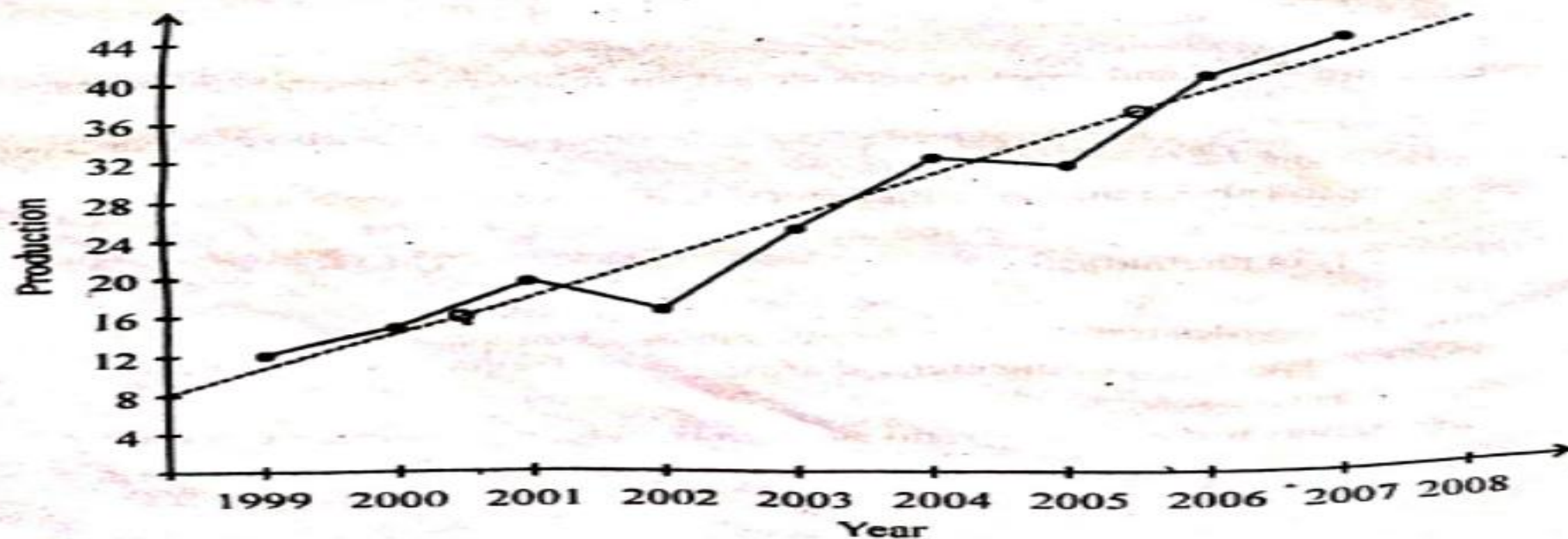
$\bar{x}_1$ = Average production for first part

$$= \frac{12 + 15 + 20 + 18}{4} = \frac{65}{4} = 16.25$$

$\bar{x}_2$ = Average production for second part

$$= \frac{32 + 30 + 40 + 44}{4} = \frac{146}{4} = 36.50$$

We plot $\bar{x}_1$ and $\bar{x}_2$ against the mid-points 2000.5 of 1999-2002 and 2005.5 of 2004-2007 respectively. The straight line obtained by joining two points represents trend line.



3. Method of Moving Averages

Moving Averages Method gives a trend with a fair degree of accuracy. In this method, we take arithmetic mean of the values for a certain time span. The time span can be three-years, four -years, five- years and so on depending on the data set and our interest. We will see the working procedure of this method.

Procedure:

- (i) Decide the period of moving averages (three- years, four -years).
- (ii) In case of odd years, averages can be obtained by calculating,

$$\frac{a+b+c}{3}, \frac{b+c+d}{3}, \frac{c+d+e}{3}, \frac{d+e+f}{3}, \dots$$

- (iii) If the moving average is an odd number, there is no problem of centering it, the average value will be centered besides the second year for every three years.

In case of even years, averages can be obtained by calculating,

$$\frac{a+b+c+d}{4}, \frac{b+c+d+e}{4}, \frac{c+d+e+f}{4}, \frac{d+e+f+g}{4}, \dots$$

(v) If the moving average is an even number, the average of first four values will be placed between 2nd and 3rd year, similarly the average of the second four values will be placed between 3rd and 4th year. These two averages will be again averaged and placed in the 3rd year. This continues for rest of the values in the problem. This process is called as centering of the averages.

Example 1: Calculate three-yearly moving averages of number of students studying in a higher secondary school in a particular village from the following data.

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Sale	332	317	357	392	402	405	410	427	435	438

Solution:

Computation of three- yearly moving averages.

Year	Number of students	3- yearly moving Total	3- yearly moving Averages
1995	332	---	---
1996	317	1006	335.33
1997	357	1066	355.33
1998	392	1151	383.67
1999	402	1199	399.67
2000	405	1217	405.67
2001	410	1242	414.00
2002	427	1272	424.00
2003	435	1300	433.33
2004	438	---	---

Example 2: Calculate four-yearly moving averages of number of students studying in a higher secondary school in a particular city from the following data.

Year	2001	2002	2003	2004	2005	2006	2007	2008
Sales	124	120	135	140	145	158	162	170

Solution:

Computation of four- yearly moving averages.

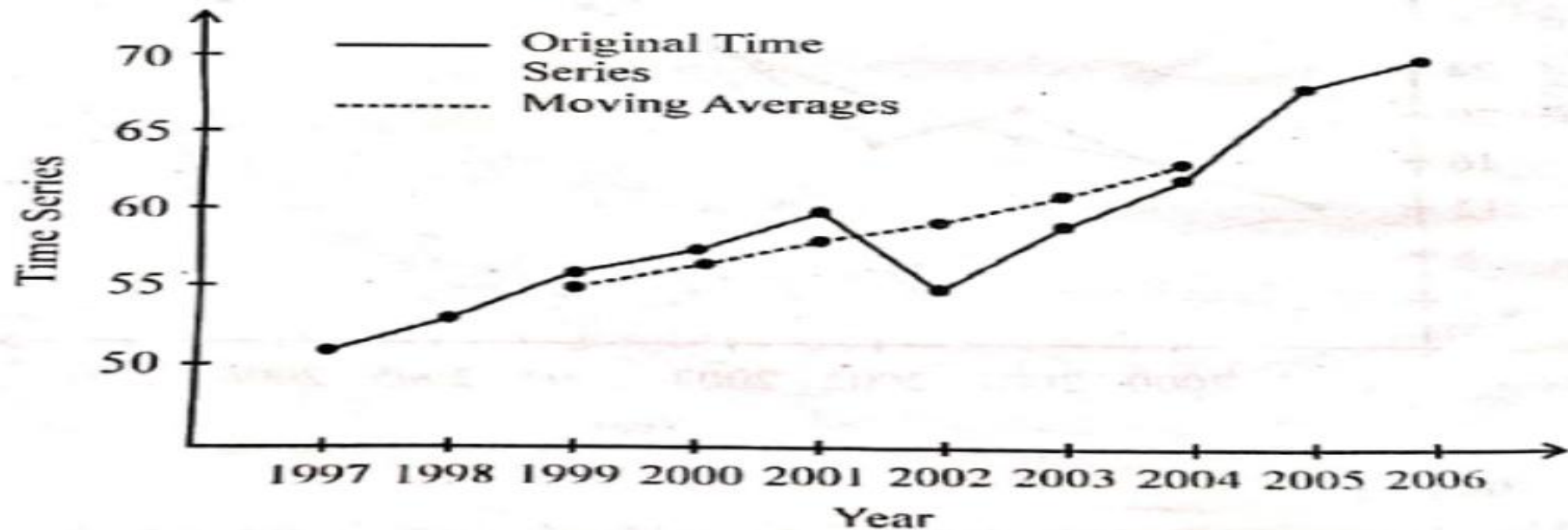
Year	Student	Total	4 yearly Moving Average	4 yearly Central Moving Average
2001	124			
2002	120			
		519	129.75	
2003	135			132.375
		540	135	
2004	140			139.75
		578	144.5	
2005	145			147.875
		605	151.25	
2006	158			155
		635	158.75	
2007	162			
2008	170			

Find 5 yearly moving averages for the following data. Plot the given data and moving averages on a graph paper

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
Sales (in lakhs of Rs)	51	53	56	57	60	55	59	62	68	70

Year	Sales (in lakhs of ₹)	5 Yearly Total (Moving Average)
1997	51	...
1998	53	...
1999	56	$(51 + 53 + 56 + 57 + 60) / 5 = 55.4$
2000	57	$(53 + 56 + 57 + 60 + 55) / 5 = 56.2$
2001	60	$(56 + 57 + 60 + 55 + 59) / 5 = 57.4$
2002	55	$(57 + 60 + 55 + 59 + 62) / 5 = 58.6$
2003	59	$(60 + 55 + 59 + 62 + 68) / 5 = 60.8$
2004	62	$(55 + 59 + 62 + 68 + 70) / 5 = 62.8$
2005	68	...
2006	70	...

The graph of Sales and Moving Averages (on y-axis) against years (on x-axis) is as follows :



4. Method of Least Squares

The line of best fit is a line from which the sum of the deviations of various points is zero. This is the best method for obtaining the trend values. It gives a convenient basis for calculating the line of best fit for the time series. It is a mathematical method for measuring trend. Further the sum of the squares of these deviations would be least when compared with other fitting methods. So, this method is known as the Method of Least Squares and satisfies the following conditions:

- (i) The sum of the deviations of the actual values of Y and $\bar{Y}$ estimated value of Y is Zero. that is $\Sigma(Y - \bar{Y}) = 0$.
- (ii) The sum of squares of the deviations of the actual values of Y and $\bar{Y}$ (estimated value of Y) is least. that is $\Sigma(Y - \bar{Y})^2$ is least ;

Procedure:

(i) The straight line trend is represented by the equation $Y = a + bX$...(1)

where Y is the actual value, X is time, a , b are constants

(ii) The constants 'a' and 'b' are estimated by solving the following two normal

Equations $\Sigma Y = n a + b \Sigma X$...(2)

$$\Sigma XY = a \Sigma X + b \Sigma X^2 \text{ ...(3)}$$

Where 'n' = number of years given in the data.

(iii) By taking the mid-point of the time as the origin, we get $\Sigma X = 0$

(iv) When $\Sigma X = 0$, the two normal equations reduces to

$$\Sigma Y = n a + b (0) \quad ; \quad a = \frac{\Sigma Y}{n} = \bar{Y}$$
$$\Sigma XY = a(0) + b \Sigma X^2 \quad ; \quad b = \frac{\Sigma XY}{\Sigma X^2}$$

The constant 'a' gives the mean of Y and 'b' gives the rate of change (slope).

(v) By substituting the values of 'a' and 'b' in the trend equation (1), we get the Line of Best Fit.

Example 1: Given below are the data relating to the production of sugarcane in a district. Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	2000	2001	2002	2003	2004	2005	2006
Prod. of Sugarcane	40	45	46	42	47	50	46

Solution:

Computation of trend values by the method of least squares (ODD Years).

Year(x)	Production of Sugarcane(Y)	X=(x-2003)	X ²	XY	Y _t =a+bX	
2000	40	-3	9	-120	0	
2001	45	-2	4	-90	0	
2002	46	-1	1	-46	0	
2003	42	0	0	0	0	
2004	47	1	1	47	0	
2005	50	2	4	100	0	
2006	46	3	9	138	0	
	ΣY=316	ΣX=0	ΣX ² =28	ΣXY=29		

$$a = \frac{\sum Y}{n} = \frac{316}{7} = 45.143 ; b = \frac{\sum XY}{\sum X^2} = \frac{29}{28} = 1.036$$

Therefore, the required equation of the straight line trend is given by

$$Y = a + bX;$$

$$Y = 45.143 + 1.036 (x - 2003)$$

The trend values can be obtained by

$$\text{When } X = 2000, Y_t = 45.143 + 1.036(2000 - 2003) = 42.035$$

$$\text{When } X = 2001, Y_t = 45.143 + 1.036(2001 - 2003) = 43.071,$$

similarly other values can be obtained.

Example 2: Given below are the data relating to the sales of a product in a district. Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	1995	1996	1997	1998	1999	2000	2001	2002
Sales	6.7	5.3	4.3	6.1	5.6	7.9	5.8	6.1

Solution: Computation of trend values by the method of least squares.

In case of EVEN number of years, let us consider

$$X = \frac{(x - \text{Arithmetic mean of two middle years})}{0.5}$$

Year(x)	Sales(Y)	$X = (x - 1998.5) / 0.5$	X^2	XY	$Y_t = a + bX$	
1995	6.7	-7	49	-46.9	5.616667	
1996	5.3	-5	25	-26.5	5.719048	
1997	4.3	-3	9	-12.9	5.821429	
1998	6.1	-1	1	-6.1	5.92381	
1999	5.6	1	1	5.6	6.02619	
2000	7.9	3	9	23.7	6.128571	
2001	5.8	5	25	29	6.230952	
2002	6.1	7	49	42.7	6.333333	
	$\sum Y = 47.8$	$\sum X = 0$	$\sum X^2 = 168$	$\sum XY = 8.6$		

$$a = \frac{\sum Y}{n} = \frac{47.8}{8} = 5.975 \quad ; \quad b = \frac{\sum XY}{\sum X^2} = \frac{4.3}{42} = 0.10238$$

Therefore, the required equation of the straight line trend is given by

$$Y = a + bX \quad ; \quad Y = 5.975 + 0.10238 X.$$

$$\text{When } X = 1995, Y_t = 5.975 + 0.10238 \left(\frac{1995 - 1998.5}{0.5} \right) = 5.6166$$

$$\text{When } X = 1996, Y_t = 5.975 + 0.10238 \left(\frac{1996 - 1998.5}{0.5} \right) = 5.7190$$

similarly other values can be obtained.

Note:

- (i) Future forecasts made by this method are based only on trend values.
- (ii) The predicted values are more reliable in this method than the other methods.

Methods of measuring Seasonal Variations By Simple Averages:

Procedure:

- (i) Arrange the data by months, quarters or years according to the data given.
- (ii) Find the sum of the each months, quarters or year.
- (iii) Find the average of each months, quarters or year.
- (iv) Find the average of averages, and it is called Grand Average (G)
- (v) Compute Seasonal Index for every season (i.e) months, quarters or year is given by

$$\text{Seasonal Index (S.I)} = \frac{\text{Seasonal Average}}{\text{Grand average}} \times 100$$

(vi) If the data is given in months

$$\text{S.I for Jan} = \frac{\text{Monthly Average (for Jan)}}{\text{Grand average}} \times 100$$

$$\text{S.I for Feb} = \frac{\text{Monthly Average (for Feb)}}{\text{Grand average}} \times 100$$

Similarly we can calculate SI for all other months.

(vii) If the data is given in quarter

$$\text{S.I for I Quarter} = \frac{\text{Average of I quarter}}{\text{Grand average}} \times 100$$

$$\text{S.I for II Quarter} = \frac{\text{Average of II quarter}}{\text{Grand average}} \times 100$$

$$\text{S.I for III Quarter} = \frac{\text{Average of III quarter}}{\text{Grand average}} \times 100$$

$$\text{S.I for IV Quarter} = \frac{\text{Average of IV quarter}}{\text{Grand average}} \times 100$$

Example 1: Calculate the seasonal index for the monthly sales of a product using the method of simple averages.

Months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Year												
2001	15	41	25	31	29	47	41	19	35	38	40	30
2002	20	21	27	19	17	25	29	31	35	39	30	44
2003	18	16	20	28	24	25	30	34	30	38	37	39

Solution:

Computation of seasonal Indices by method of simple averages.

months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Years												
2001	15	41	25	31	29	47	41	19	35	38	40	30
2002	20	21	27	19	17	25	29	31	35	39	30	44
2003	18	16	20	28	24	25	30	34	30	38	37	39
Monthly Total	53	78	72	78	70	97	100	84	100	115	107	113
Monthly Averages	17.67	26	24	26	23.33	32.33	33.33	28	33.33	38.33	35.67	37.67

$$\text{S.I for Jan} = \frac{\text{Monthly Average (for Jan)}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{355.582}{12} = 29.63$$

$$\text{S.I for Jan} = \frac{17.666}{29.361} \times 100 = 59.62 ; \quad \text{S.I for Feb} = \frac{26}{29.361} \times 100 = 87.77 ;$$

Similarly other seasonal index values can be obtained.

Months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Seasonal Index	59.62	87.77	80.99	87.77	78.74	109.12	112.49	94.49	112.49	129.36	120.36	126.89

Example 2: Calculate the seasonal index for the quarterly production of a product using the method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400

Solution:

Computation of seasonal Indices by method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400
Quarterly Total	1463	1872	2170	2321
Quarterly Averages	243.83	312	361.67	386.83

$$\text{S.I for I Quarter} = \frac{\text{Average of I quarter}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{1304.333}{4} = 326.0833$$

$$\text{S.I for I Q} = \frac{243.8333}{326.0833} \times 100 = 74.77 ;$$

$$\text{S.I for II Q} = \frac{312}{326.0833} \times 100 = 95.68 ;$$

$$\text{S.I for III Q} = \frac{361.6667}{326.0833} \times 100 = 110.91 ;$$

$$\text{S.I for IV Q} = \frac{386.833}{326.0833} \times 100 = 118.63$$